

Alexandria National University  
Faculty of Computers and Data Science  
Cyber Security Program



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## Log File Analysis with Bash Script Report

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Course:

Information Security Management

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## Introduction

This is a report of web server access log analysis. The log was analyzed using a Bash script to determine the key findings such as trends in requests, distribution of status codes, user behavior, and failure trends. The objective is to recognize the likely problems and recommend areas of improvement in system security and reliability

## Process

Firstly I got the Log File From Internet to Analyze it by the bash Script

### Log File Sample :

**127.0.0.1 - - [08/May/2025:14:56:35 +0000] "GET /index.html HTTP/1.1" 200 1024**

- **127.0.0.1**            The IP address of the client making the request
- **- -**                    User identifier fields (unused in this case, shown as -)
- **[08/May/2025:14:56:35 +0000]**            Date and time of the request (8 May 2025, 14:56:35 UTC)
- **"GET /index.html HTTP/1.1"**            The HTTP request method (GET), resource (/index.html), protocol (HTTP/1.1)
- **200**                    HTTP status code (success)
- **1024**                    Size of the response in bytes

## After That I made The Bash Script File Which Will Analyze the Log File :

```
$ log_analysis.sh X
C:\Users\OMAR\Desktop> Task 3 > $ log_analysis.sh
1 |#!/bin/bash
2
3 LOG_FILE="access.log" # Replace with your log file path
4
5 # 1. Request Counts
6 total_requests=$(wc -l < "$LOG_FILE")
7 get_requests=$(grep "GET" "$LOG_FILE" | wc -l)
8 post_requests=$(grep "POST" "$LOG_FILE" | wc -l)
9
10 # 2. Failures (4xx or 5xx)
11 failures=$(awk '$9 ~ /^4|^5/' "$LOG_FILE" | wc -l)
12 failure_percent=$(awk -v f="$failures" -v t="$total_requests" 'BEGIN { printf "%.2f", (f/t)*100 }')
13
14 # 3. Unique IP Addresses and Request Methods by IP
15 unique_ips=$(awk '{print $1}' "$LOG_FILE" | sort | uniq)
16 ip_requests=$(awk '{print $1, $6}' "$LOG_FILE" | sort | uniq -c)
17
18 # 4. Top User (Most Active IP)
19 most_active_ip=$(awk '{print $1}' "$LOG_FILE" | sort | uniq -c | sort -nr | head -n 1)
20
21 # 5. Daily Request Averages
22 requests_by_day=$(awk '{print $4}' "$LOG_FILE" | cut -d: -f1 | sort | uniq -c)
23 total_days=$(echo "$requests_by_day" | wc -l)
24 daily_average=$(echo "scale=2; $total_requests / $total_days" | bc)
25
26 # 6. Failure Analysis (Highest Failure Requests by Day)
27 failures_by_day=$(awk '$9 ~ /^4|^5/ {print $4}' "$LOG_FILE" | cut -d: -f1 | sort | uniq -c | sort -nr)
28
29 # 7. Requests by Hour
30 requests_by_hour=$(awk '{print $4}' "$LOG_FILE" | cut -d: -f2 | sort | uniq -c)
31
32 # 8. Status Code Breakdown
33 status_breakdown=$(awk '{print $9}' "$LOG_FILE" | sort | uniq -c | sort -nr)
34
35 # 9. Most Active User by Method (GET and POST Requests)
36 get_requests_by_ip=$(grep "GET" "$LOG_FILE" | awk '{print $1}' | sort | uniq -c | sort -nr)
37 post_requests_by_ip=$(grep "POST" "$LOG_FILE" | awk '{print $1}' | sort | uniq -c | sort -nr)
38
39 # 10. Patterns in Failure Requests (Failure Counts by Hour)
40 failure_patterns=$(awk '$9 ~ /^4|^5/ {print $4}' "$LOG_FILE" | cut -d: -f2 | sort | uniq -c | sort -nr)
41
42 # Save data for Python visualization
43 echo "total_requests,get_requests,post_requests,failures,failure_percent" > data.csv
44 echo "$total_requests,$get_requests,$post_requests,$failures,$failure_percent" >> data.csv
45
46 # Add breakdown data into CSV files with headers
47 echo "Hour,Requests" > requests_by_hour.csv
48 echo "$requests_by_hour" >> requests_by_hour.csv
49
50 echo "Status Code,Count" > status_breakdown.csv
51 echo "$status_breakdown" >> status_breakdown.csv
52
53 echo "IP,GET Requests" > get_requests_by_ip.csv
54 echo "$get_requests_by_ip" >> get_requests_by_ip.csv
55
56 echo "IP,POST Requests" > post_requests_by_ip.csv
57 echo "$post_requests_by_ip" >> post_requests_by_ip.csv
```

```

58
59 echo "Day,Requests" > requests_by_day.csv
60 echo "$requests_by_day" >> requests_by_day.csv
61
62 echo "Day,Failures" > failures_by_day.csv
63 echo "$failures_by_day" >> failures_by_day.csv
64
65 echo "Hour,Failures" > failure_patterns.csv
66 echo "$failure_patterns" >> failure_patterns.csv
67
68 # Output results to the console
69 echo "==== Log File Analysis ====="

```

```

70 echo "Total Requests: $total_requests"
71 echo "GET Requests: $get_requests"
72 echo "POST Requests: $post_requests"
73 echo ""
74 echo "Unique IPs: $(echo "$unique_ips" | wc -l)"
75 echo "Request Count per IP:"
76 echo "$ip_requests"
77 echo ""
78 echo "Failed Requests: $failures"
79 echo "Failure Percentage: $failure_percent%"
80 echo ""
81 echo "Most Active IP:"
82 echo "$most_active_ip"
83 echo ""
84 echo "Days in Log: $total_days"
85 echo "Average Requests/Day: $daily_average"
86 echo ""
87 echo "Days with Most Failures:"
88 echo "$failures_by_day"
89 echo ""
90 echo "Requests Per Hour:"
91 echo "$requests_by_hour"
92 echo ""
93 echo "Daily Request Trend:"
94 echo "$requests_by_day"
95 echo ""
96 echo "Status Code Breakdown:"
97 echo "$status_breakdown"
98 echo ""
99 echo "Top GET IP:"
100 echo "$get_requests_by_ip"
101 echo "Top POST IP:"
102 echo "$post_requests_by_ip"

```

```

103 echo ""
104 echo "Failure Requests by Hour:"
105 echo "$failure_patterns"

```

**log\_analysis.sh**, is designed to perform automated analysis of a web server access log file. It extracts key statistics & saves summary data for reporting and visualization purposes.

The script performs the following functions:

#### **Request Counts:**

- Counts the total number of requests in the log.
- Counts how many of those requests are GET and POST methods.

#### **Failure Requests:**

- Identifies requests that resulted in client or server errors (status codes 4xx and 5xx).
- Calculates the percentage of failed requests relative to total requests.

#### **Unique IP Addresses:**

- Lists unique IP addresses that accessed the server.
- Counts GET and POST requests per IP address.

#### **Top User Identification:**

- Determines the most active IP address (the one with the highest number of requests).

#### **Daily Request Analysis:**

- Counts the number of requests per day.
- Calculates the average number of requests per day.

#### **Failure Analysis by Day:**

- Identifies the days with the highest number of failed requests.

## Request Patterns by Hour:

- Counts the number of requests made during each hour of the day.

## Status Code Breakdown:

- Provides a summary of HTTP status codes (e.g., 200, 404, 500) and their respective frequencies.

## Most Active IP by Method:

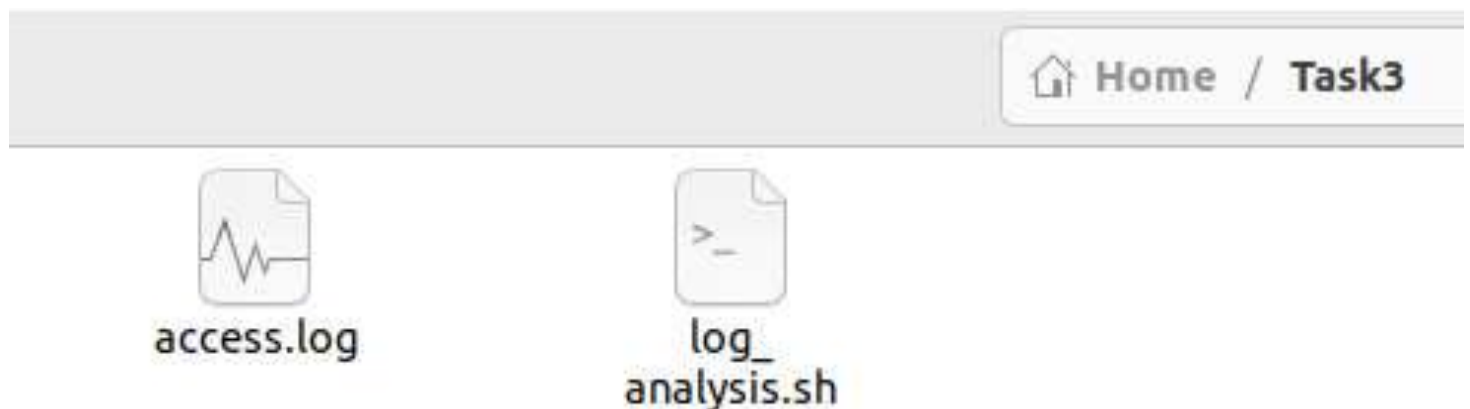
- Identifies which IP made the most GET and POST requests.

## Failure Patterns by Hour:

- Analyzes the distribution of failed requests across different hours.

So I Opened my Ubuntu Virtual Machine to test the Two Files Outcome

So I make A Directory and Added the Two Files in it



Then I Opened the Directory in Terminal to test its Run :

```
omar@omar-virtual-machine: ~/Task3
omar@omar-virtual-machine:~$ pwd
/home/omar
omar@omar-virtual-machine:~$ cd Task3
omar@omar-virtual-machine:~/Task3$ ls
access.log  log_analysis.sh
```

After That This Output Appeared for me :

```
omar@omar-virtual-machine:~/Task3$ chmod +x log_analysis.sh
omar@omar-virtual-machine:~/Task3$ ./log_analysis.sh
==== Log File Analysis ====
Total Requests: 832
GET Requests: 499
POST Requests: 333

Unique IPs: 3
Request Count per IP:
    167 127.0.0.1 "GET
    167 127.0.0.1 "POST
    166 192.168.0.1 "GET
    166 192.168.0.1 "POST
    166 192.168.0.2 "GET

Failed Requests: 332
Failure Percentage: 39.90%

Most Active IP:
    334 127.0.0.1
```

```
Days in Log: 1
Average Requests/Day: 832.00
```

```
Days with Most Failures:
  332 [08/May/2025
```

```
Requests Per Hour:
  666 14
  166 15
```

```
Daily Request Trend:
  832 [08/May/2025
```

```
Status Code Breakdown:
  499 200
  166 500
  166 404
    1
```

```
Top GET IP:
  167 127.0.0.1
  166 192.168.0.2
  166 192.168.0.1
```

```
Top POST IP:
  167 127.0.0.1
  166 192.168.0.1
```

```
Failure Requests by Hour:
  332 14
```

```
omar@omar-virtual-machine:~/Task3$
```

The Bash script **log\_analysis.sh** was made **executable** and successfully executed on the **access.log** file. The script processed the log and generated a detailed statistical analysis of the server requests.

The Output included:

- **Total requests analyzed:** 832
- **GET requests:** 499
- **POST requests:** 333
- **Unique IP addresses:** 3
- **Most active IP address:** 127.0.0.1 with 334 requests
- **Failed requests (4xx/5xx):** 332 (39.90% failure rate)
- **Top GET IP:** 127.0.0.1
- **Top POST IP:** 127.0.0.1
- **Day with highest failures:** 8th May 2025
- **Requests distributed** mostly at 2 PM (666 requests at hour 14)
- **Status code breakdown:**

**200 OK:** 499

**500 Internal Server Error:** 166

**404 Not Found:** 1

CSV files were also generated for further analysis or visualization:

